

Synthetic Data Generation and Cross-Modal Modeling of Time-Series Data for Accurate and Robust Health Monitoring

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Abstract. Wearable and physiological sensors provide continuous health-care monitoring, but real-world time series are frequently limited, unbalanced, and noisy, reducing robustness in safety-critical activities such as fall detection and seizure monitoring. My PhD research focuses on how generative models and representation learning can (i) synthesize realistic, task-useful signals, (ii) quantify rare events under prior probability shift, and (iii) connect raw sensors to structured motion for improved interpretability. I developed TransConv-DDPM for diffusion-based time-series generation [2, 4], CC-GMNet-TS for class-conditioned prevalence estimation [3], and a sensor-to-skeleton latent diffusion model with graph attention for cross-modal fall detection. Results on SmartFallMM, EEG, EMG, and UCI-HAR show gains in realism and downstream utility. Future work will combine these ideas into a robust, interpretable, privacy-aware framework.

Keywords: Time-series generation · Diffusion models · Quantification
· Fall detection · Cross-modal learning.

1 Introduction and Research Questions

Wearable inertial and physiological sensors enable continuous monitoring of falls, mobility, and neurological events. However, building large, diverse, well-labeled healthcare time-series datasets is difficult due to privacy, safety, cost, and collection constraints. As a result, available datasets are often small, noisy, and imbalanced, with rare but critical events such as falls or seizures appearing only sparsely. These limitations increase subject-specific variation and distribution shift between training and deployment, reducing the reliability of supervised deep models.

Another challenge is that raw accelerometer or EEG signals provide limited structural information about underlying body motion or physiological processes. This makes it harder to distinguish rare clinical events from similar activities of daily living (ADLs), especially under data scarcity and prior probability shift.

My overarching research question is:

How can we design generative and cross-modal representation-learning methods that synthesize realistic, task-useful time-series and connect low-level sensor streams with higher-level structure to improve robustness and interpretability in fall detection and related healthcare applications?

This thesis studies three sub-questions:

1. How can diffusion-based generators model multi-scale dynamics and long-range temporal dependencies in physiological and biomechanical signals, and how much can synthetic samples improve downstream performance?
2. How can we perform *quantification* for imbalanced time-series under prior shift using class-conditioned latent models?
3. How can wearable inertial signals be mapped to structured skeleton motion using graph-aware diffusion and attention to improve interpretability and fall vs. ADL discrimination?

These questions progress from data generation, to prevalence estimation, to cross-modal structured representations for more robust and interpretable fall detection.

2 Significance, Challenges, and Related Work

Accurate fall detection and rare-event prevalence estimation are important for elderly care, rehabilitation, and neurological monitoring: false negatives risk missed intervention, while false positives increase alarm fatigue. Synthetic data that augments scarce datasets, and representations that make rare events more separable, have the potential to increase safety and scalability.

Previous work on time-series generation includes GANs, VAEs, and diffusion models; diffusion often gives higher fidelity but commonly uses convolutional backbones that underuse long-range context [1]. Quantification methods such as Classify-and-Count adjust to new priors but rely on calibrated instance scores; deep quantifiers (e.g., QuaNet, GMNet) are less explored for multimodal time-series. In fall detection, CNN/LSTM models operate on inertial streams, while skeleton-based GNNs (e.g., ST-GCN) perform well but usually require cameras, raising privacy constraints. Sensor-to-pose work prioritizes motion capture accuracy over cross-modal creation designed for fall detection.

3 Completed and Ongoing Work

3.1 TransConv-DDPM: Enhanced Diffusion for Time-Series Generation

In my COMPSAC 2025 and IEEE PULSE 2025 work, I proposed *TransConv-DDPM*, a diffusion model for biomechanical and physiological time-series [2, 4]. It augments U-Net diffusion with multi-scale convolutions and a bottleneck transformer. On SmartFallMM, CHB-MIT EEG, and stick-balancing, it outperformed TimeGAN/Diffusion-TS and improved fall detection when used for synthetic augmentation.

3.2 CC-GMNet-TS: Class-Conditioned Quantification for Imbalanced Time-Series

My PAKDD 2026 paper proposes *CC-GMNet-TS*, a deep quantification model that estimates bag-level class prevalences under prior shift [3]. It uses a transformer encoder and class-conditioned Gaussian mixtures in a bounded latent space. On EMG, SmartFallMM, and UCI-HAR, it outperformed classical and deep baselines, especially under severe imbalance.

3.3 Sensor-to-Skeleton Diffusion with Graph Attention

In ongoing work, I am building a *sensor-to-skeleton latent diffusion* model that generates interpretable skeleton sequences from wearable accelerometer windows for fall vs. ADL classification without cameras. It uses sensor conditioning, a joint-aligned graph latent space, graph-aware diffusion with temporal/joint attention, and a transformer classifier. Early results on SmartFallMM and UP-Fall show accuracy/F1 gains.

4 Proposed Research Plan and Timeline

Next, I will: (i) develop a diffusion-based framework for robust time-series imputation on multivariate wearable data (SmartFallMM, UP-Fall, KFall), evaluating reconstruction quality and impact on fall/activity classification; (ii) unify generation and quantification by using synthetic data to probe prevalence estimation under controlled prior shift and exploring quantification-aware generative objectives; (iii) extend cross-modal sensor-to-skeleton modeling to additional modalities (e.g., IMU+EMG) with stronger physical and kinematic priors; and (iv) study robustness and privacy under subject/device shift, label noise, and missing channels, deriving evaluation protocols that link generative/imputation quality to downstream utility. Over the next 6-9 months, I plan to improve TransConv-DDPM with robustness analyses, implement the imputation framework, and pursue a journal version. In the next 9-12 months, I plan to scale multi-modal sensor-to-skeleton diffusion, improve low-step sampling for near real-time inference, and integrate imputation and generation into unified pipelines. The final year will consolidate results and complete dissertation writing and defense.

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